

1 test 1

$$\begin{aligned}\int p dx &= n\hbar \quad p = m\dot{x}; \quad E = \frac{1}{2}m\omega^2 a^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\omega^2 x^2 \\ \dot{x} &= \omega\sqrt{a^2 - x^2}; \quad 2 \int_{-a}^a m\omega\sqrt{a^2 - x^2} dx = n\hbar \\ 2 \int_{-a}^a m\omega a^2 \sqrt{1 - \frac{x^2}{a^2}} d\frac{x}{a} &= 2m\omega a^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos\theta d\sin\theta = m\omega a^2 \pi = n\hbar \rightarrow a^2 = \frac{n\hbar}{m\omega\pi} \\ E_n &= \frac{\omega n\hbar}{2\pi}\end{aligned}$$

2 test 2

$$\begin{aligned}\psi &= \frac{1}{\sqrt{2}}(1 + \cos 2kx) \\ \psi(k') &= \frac{1}{\sqrt{2\pi}} \int \frac{1}{\sqrt{2}}(1 + \cos 2kx)e^{-ik'x} dx \\ &\Rightarrow \frac{1}{2\sqrt{\pi}}(\sqrt{2\pi}\delta(k') + \frac{1}{2}(\sqrt{2\pi}\delta(k' - 2k) + \sqrt{2\pi}\delta(k' + 2k))) \\ p_1 &= 0, P = \frac{2}{3}; p_2 = 2k\hbar, P = \frac{1}{6}; p_3 = -2k\hbar, P = \frac{1}{6} \\ \overline{E} &= \frac{4k^2\hbar^2}{2m} \frac{1}{6} \times 2 = \frac{2k^2\hbar^2}{3m}\end{aligned}$$

3 test 3

$$\begin{aligned}\psi(0) &= \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2\pi\hbar}}e^{ipx/\hbar} + \frac{1}{\sqrt{2\pi\hbar}}e^{-ipx/\hbar}\right) \\ \psi_{E1} &= e^{ipx/\hbar}, \psi_{E2} = e^{-ipx/\hbar}; E_1 = E_2 = \frac{p^2}{2m} \\ \psi(t) &= \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2\pi\hbar}}e^{-ipx/\hbar - iEt/\hbar} + \frac{1}{\sqrt{2\pi\hbar}}e^{-ipx/\hbar - iEt/\hbar}\right), E = \frac{p^2}{2m}\end{aligned}$$

4 test 4

$$\psi_E^* \psi_{E'} = \delta_{EE'}$$

PROOF:

$$\begin{aligned}
\hat{H}\psi_E &= E\psi_E \rightarrow \hat{H}^*\psi_E^* = E\psi_E^* \\
\hat{H}\psi_{E'} &= E'\psi_{E'} \\
\psi_E^*\hat{H}\psi_{E'} &= \psi_E^*E'\psi_{E'} \\
\hat{H}^*\psi_E^*\psi_{E'} &= E\psi_E^*\psi_{E'} \\
\Rightarrow 0 &= (E' - E)\psi_E^*\psi_{E'} \Rightarrow \psi_E^*\psi_{E'} = \delta_{EE'}
\end{aligned}$$

5 test 5

$$V = \begin{cases} \infty & x < 0 \\ 0 & 0 < x < a \\ V_0 & x > a \end{cases}$$

SOLVE

$$\begin{aligned}
-\frac{\hbar^2}{2m}\nabla^2\psi - E\psi &= 0 \quad 0 < x < a \\
\Rightarrow \psi &= A\sin\sqrt{\frac{2mE}{\hbar^2}}x = A\sin kx \quad 0 < x < a \\
-\frac{\hbar^2}{2m}\nabla^2\psi - (E - V_0)\psi &= 0 \quad x > a, k = \sqrt{\frac{2mE}{\hbar^2}} \\
\Rightarrow \psi &= Be^{-i\sqrt{\frac{2m(E-V_0)}{\hbar^2}}x} = Be^{-k'x} \quad x > a, k' = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \\
A\sin ka &= Be^{-k'a} \\
\frac{d}{dx}\ln(A\sin kx)|_{x=a} &= \frac{d}{dx}\ln(Be^{-k'x})|_{x=a} \\
\Rightarrow k\cos kx/\sin kx|_{x=a} &= -k' \rightarrow \cot ka = -\frac{k'}{k} \\
k'^2 + k^2 &= \frac{2mV_0}{\hbar^2} \\
a\sqrt{\frac{2mV_0}{\hbar^2}} > \frac{\pi}{2} \Rightarrow V > \frac{\pi^2}{4}\hbar^2\frac{1}{2ma^2} &= \frac{\pi^2\hbar^2}{8ma^2}
\end{aligned}$$

6 test 6

$$[x, \hat{p}^2] = ?$$

$$[x, \hat{p}]\hat{p} + \hat{p}[x, \hat{p}] = 2i\hbar$$

7 test 7

$$[A, B] = C, [B, C] = 0; [A, B^n] = ?, [A, e^{\lambda B}] = ?$$

$$[A, B^2] = CB + BC = 2CB; [A, B^3] = 3CB^2$$

$$\text{set } [A, B^{n-1}] = (n-1)CB^{n-2} \text{ for any } n$$

$$[A, B^n] = B^{n-1}[A, B] + [A, B^{n-1}]B = CB^{n-1} + (n-1)CB^{n-1} = nCB^{n-1}$$

$$[A, e^{\lambda B}] = \sum_n \frac{\lambda^n}{n!} [A, B^n] = \sum_n \frac{\lambda^n}{n!} nCB^{n-1} = C \frac{\partial}{\partial B} e^{\lambda B}$$

8 test 8

$$\psi = \frac{1}{\sqrt{2}}(Y_{1-1} + Y_{10}) \quad \bar{l}_z = ?$$

$$l_z \psi = \frac{1}{\sqrt{2}}(-\hbar Y_{1-1} + 0)$$

$$\psi^* l_z \psi = -\frac{1}{2}\hbar$$

9 test 9

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad v_i = ?, \lambda_i = ?, \psi_0 = \chi_{\frac{1}{2}} = Av_1 + Bv_2, A = ?, B = ?$$

$$\begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 1, \lambda_2 = -1$$

$$v_1 = \frac{1}{\sqrt{2}}\chi_1 + \frac{1}{\sqrt{2}}\chi_0, v_2 = \frac{1}{\sqrt{2}}\chi_1 - \frac{1}{\sqrt{2}}\chi_0$$

$$\psi_0 = \frac{1}{\sqrt{2}}v_1 + \frac{1}{\sqrt{2}}v_2$$

10 test 10

$$\psi_0 = \frac{1}{2}Y_{10} + \frac{\sqrt{3}}{2}Y_{11}, \bar{l}_x = ?$$

$$\begin{aligned}
l_x(\frac{1}{2}Y_{10} + \frac{\sqrt{3}}{2}Y_{11}) &= \frac{1}{2}(l_+ + l_-)(\frac{1}{2}Y_{10} + \frac{\sqrt{3}}{2}Y_{11}) \\
&\Rightarrow \frac{1}{4}(a_{10}Y_{11} + b_{10}Y_{1-1}) + \frac{\sqrt{3}}{4}(b_{11}Y_{10}) \\
\bar{l}_x &= \frac{\sqrt{3}}{8}b_{11} + \frac{\sqrt{3}}{8}a_{10} = \frac{\sqrt{6}}{4}
\end{aligned}$$

11 Sep 18 HW

straight line wave, ring wave, 1.1, 1.2, 1.3, 1.4 P8: 1, 2, 5

I:

$$\begin{aligned}
n\lambda = 2a, p &= \frac{\hbar}{\lambda} \Rightarrow p = \frac{n\hbar}{2a} \\
E &= \frac{n^2\hbar^2}{8ma^2}
\end{aligned}$$

II:

$$\begin{aligned}
n\lambda = 2\pi r, p &= \frac{\hbar}{\lambda} \Rightarrow p = \frac{n\hbar}{2\pi r} \\
E &= \frac{n^2\hbar^2}{8\pi^2 r^2 m}
\end{aligned}$$

III: 1.1

(1)

$$\begin{aligned}
\bar{E} &= 1/2m \int \psi^* \hat{P} \hat{P} \psi + \psi^* V \psi dx^3 = 1/2m \int (\hat{P}\psi)^* \hat{P}\psi + \psi^* V \psi dx^3, \\
\omega &= i\hbar \nabla \psi^* (-i\hbar \nabla) \psi + \psi^* V \psi = \hbar^2/2m \nabla \psi^* \cdot \nabla \psi + \psi^* V \psi
\end{aligned}$$

(2)

$$\frac{\partial}{\partial t} \omega + \nabla \cdot s = 0$$

$$\begin{aligned}
\frac{\partial}{\partial t}\omega &= \frac{\hbar^2}{2m}(\nabla \frac{\partial}{\partial t}\psi^*) \cdot \nabla \psi + \frac{\hbar^2}{2m}\nabla \psi^* \cdot \nabla \frac{\partial}{\partial t}\psi + \frac{\partial}{\partial t}\psi^* V \psi + \psi^* V \frac{\partial}{\partial t}\psi \\
\nabla \cdot s &= -\frac{\hbar^2}{2m}(\nabla \frac{\partial}{\partial t}\psi^* \cdot \nabla \psi + \frac{\partial}{\partial t}\psi^* \nabla^2 \psi + \nabla \frac{\partial}{\partial t}\psi \cdot \nabla \psi^* + \frac{\partial}{\partial t}\psi \nabla^2 \psi^*) \\
&\quad - \frac{\hbar^2}{2m}(\frac{\partial}{\partial t}\psi^* \nabla^2 \psi + \frac{\partial}{\partial t}\psi \nabla^2 \psi^*) + \frac{\partial}{\partial t}\psi^* V \psi + \psi^* V \frac{\partial}{\partial t}\psi \\
&\quad \frac{\partial}{\partial t}\psi^*(E\psi - V\psi) + \frac{\partial}{\partial t}\psi(E\psi^* - V\psi^*) + \frac{\partial}{\partial t}(\psi^* V \psi) \\
&\Rightarrow \frac{\partial}{\partial t}(\psi^* E \psi) = 0
\end{aligned}$$

IV:1.2

(a)

$$\begin{aligned}
i\hbar\psi^* \frac{\partial}{\partial t}\psi &= -\frac{\hbar^2}{2m}\psi^* \nabla^2 \psi + \psi^*[V_1 + iV_2]\psi \\
-i\hbar\psi \frac{\partial}{\partial t}\psi^* &= -\frac{\hbar^2}{2m}\psi \nabla^2 \psi^* + \psi[V_1 - iV_2]\psi^* \\
i\hbar\psi^* \frac{\partial}{\partial t}\psi + i\hbar\psi \frac{\partial}{\partial t}\psi^* &= -\frac{\hbar^2}{2m}[\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*] + 2iV_2\psi^*\psi \\
i\hbar \frac{\partial}{\partial t}(\psi^*\psi) &= -\frac{\hbar^2}{2m}\nabla[\psi^* \nabla \psi - \psi \nabla \psi^*] + 2iV_2\psi^*\psi
\end{aligned}$$

(b)

$$\int \frac{\partial}{\partial t}\psi^*\psi = -\frac{\hbar}{2im} \int_S \psi^* \nabla \psi - \psi \nabla \psi^* + 2i \int V_2 \psi^*\psi$$

V:1.3

(a)

$$\hat{H}\psi = -\hat{p}^2/2m\psi = \frac{p^2}{2m}\psi = E\psi$$

(b)

$$\begin{aligned}
\psi(0) &= \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \\
\psi(t) &= \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar - i\frac{p^2 t}{2m}/\hbar}
\end{aligned}$$

(c)

$$\begin{aligned}
\psi(x) = \delta(x) &= \frac{1}{\sqrt{2\pi}} \int e^{ikx} dk \\
\hat{H}\psi &= -\frac{\hbar^2}{2m} \frac{1}{\sqrt{2\pi}} \int (-k^2) e^{ikx} dk \neq E\psi
\end{aligned}$$

(d)

$$\begin{aligned}
\psi(x) &= \frac{1}{\sqrt{2\pi}} \int e^{ikx} dk \\
\psi(x, t) &= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2\pi}} \int e^{ikx} e^{-i\frac{k^2 \hbar t}{2m}} dk \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2\pi}} \int e^{i(kx - \frac{k^2 \hbar t}{2m})} dk \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2\pi}} \int \frac{1}{-\sqrt{\frac{\hbar t}{2m}}} e^{i\frac{x^2 m}{2\hbar t}} e^{i\xi^2} d\xi \\
&= \frac{-1}{\sqrt{\frac{2\pi \hbar t}{m}}} e^{i\frac{x^2 m}{2\hbar t}} e^{i\pi/4}
\end{aligned}$$

VI:1.4

$$\psi(x, 0) = e^{ip_0 x/\hbar} \frac{1}{(\pi\alpha^2)^{1/4}} e^{-x^2/2\alpha^2}$$

(a)

$$\psi(x, 0) = e^{ip_0 x/\hbar} \frac{1}{(\pi\alpha^2)^{1/4}} e^{-x^2/2\alpha^2}$$

$$\bar{x} = \frac{1}{\sqrt{\pi\alpha^2}} \int x e^{-x^2/\alpha^2} dx = 0$$

$$\frac{1}{\sqrt{\pi\alpha^2}} \int e^{-ip_0 x/\hbar} e^{-x^2/2\alpha^2} (-i\hbar) \nabla e^{ip_0 x/\hbar} e^{-x^2/2\alpha^2} dx$$

$$\frac{1}{\sqrt{\pi\alpha^2}} \int (-i\hbar)(ip_0/\hbar - x/\alpha^2) e^{-x^2/\alpha^2} dx$$

$$= \frac{1}{\sqrt{\pi\alpha^2}} p_0 \sqrt{\pi} \alpha = p_0$$

$$\bar{x^2} = \frac{1}{\sqrt{\pi\alpha^2}} \int x^2 e^{-x^2/\alpha^2} dx = \frac{1}{\sqrt{\pi\alpha^2}} \frac{\alpha^2}{2} \sqrt{\pi} \alpha = \frac{\alpha^2}{2}$$

$$\begin{aligned}
\overline{\hat{p}^2} &= \frac{1}{\sqrt{\pi\alpha^2}} \int e^{-ip_0x/\hbar} e^{-x^2/2\alpha^2} (-\hbar^2) \nabla^2 e^{ip_0x/\hbar - x^2/2\alpha^2} dx \\
&= \frac{-\hbar^2}{P\sqrt{\pi\alpha^2}} \int e^{-ip_0x/\hbar - x^2/2\alpha^2} \nabla (ip_0/\hbar - x/\alpha^2) e^{ip_0x/\hbar - x^2/2\alpha^2} \\
&= \frac{-\hbar^2}{\sqrt{\pi\alpha^2}} \int e^{-x^2/\alpha^2} (-1/\alpha^2) dx - \frac{\hbar^2}{\sqrt{\pi\alpha^2}} \int e^{-x^2/\alpha^2} (ip_0/\hbar - x/\alpha^2)^2 dx \\
&= \frac{-\hbar^2}{\sqrt{\pi\alpha^2}} \left[-\frac{1}{\alpha^2} \sqrt{\pi}\alpha + \left(-\frac{p_0^2}{\hbar^2}\right) \sqrt{\pi}\alpha + \frac{1}{\alpha^4} \frac{\alpha^2}{2} \sqrt{\pi}\alpha \right] \\
&= \frac{\hbar^2}{\alpha^2} + p_0^2 - \frac{\hbar^2}{2\alpha^2} = \frac{\hbar^2}{2\alpha^2} + p_0^2 \\
\Delta x &= \frac{\alpha}{\sqrt{2}} \\
\Delta p &= \frac{\hbar}{\sqrt{2}\alpha} \\
\Delta x \Delta p &= \frac{\hbar}{2}
\end{aligned}$$

VII:P8.1

$$\psi(x) = Ae^{-\alpha^2 x^2/2}. A = ?$$

$$A^2 \int e^{-\alpha^2 x^2} dx = 1 = A^2 \frac{\sqrt{\pi}}{\alpha} \Rightarrow A = \sqrt{\frac{\alpha}{\sqrt{\pi}}}$$

VIII:P8.2

$$\psi^* \psi = 1$$

IX:P8.5

(a)

$$P = \int \psi^* \psi \sin\theta d\theta d\phi \cdot r dr$$

(b)

$$P = \int \psi^* \psi r dr \cdot \sin\theta d\theta d\phi$$

12 Chapter 2 HWs

2.2

$$\begin{aligned}
 & -\frac{\hbar^2}{2m}\nabla^2\psi - E\psi = 0 \\
 & \psi = A\sin\sqrt{\frac{2mE}{\hbar^2}}x, \sqrt{\frac{2mE}{\hbar^2}} = \frac{n\pi}{a} \\
 & 1 = A^2 \int_0^a \sin^2 \frac{n\pi x}{a} dx = A^2 \int_0^a \frac{1}{2} - \frac{1}{2}\cos\frac{2n\pi x}{a} dx \\
 & 1 = A^2 \frac{a}{2} \Rightarrow A = \sqrt{\frac{2}{a}} \Rightarrow \psi = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} \\
 & \bar{x} = \int_0^a \frac{2}{a} x \sin^2 \frac{n\pi x}{a} dx = \int_0^a \frac{1}{a} x - \frac{1}{a} x \cos \frac{2n\pi x}{a} dx \\
 & = \frac{a}{2} \\
 & \overline{x^2} = \int_0^a x^2 \frac{2}{a} \sin^2 \frac{n\pi x}{a} dx = \frac{1}{a} \int_0^a x^2 - x^2 \cos \frac{2n\pi x}{a} dx \\
 & = \frac{a^2}{3} - \frac{1}{a} \int_0^a x^2 \cos \frac{2n\pi x}{a} dx \\
 & = \frac{a^2}{3} - \frac{1}{a} \frac{a}{2n\pi} x^2 \sin \frac{2n\pi x}{a} + \frac{1}{2n\pi} \int_0^a 2x \sin \frac{2n\pi x}{a} dx \\
 & = \frac{a^2}{3} + \frac{1}{n\pi} \int_0^a x \sin \frac{2n\pi x}{a} dx \\
 & = \frac{a^2}{3} - \frac{a}{2n^2\pi^2} \int_0^a x d\cos \frac{2n\pi x}{a} = \frac{a^2}{3} - \frac{a^2}{2n^2\pi^2} + 0 \\
 & \Delta x^2 = \frac{a^2}{3} - \frac{a^2}{2n^2\pi^2} - \frac{a^2}{4} = \frac{a^2}{12} - \frac{a^2}{2n^2\pi^2}
 \end{aligned}$$

2.3

$$\begin{aligned}
\psi &= \sqrt{\frac{2}{a}} \cos \frac{\pi x}{a} \\
\psi(p) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\frac{a}{2}}^{\frac{a}{2}} \sqrt{\frac{2}{a}} \cos \frac{\pi x}{a} e^{-ipx/\hbar} dx \\
&= \frac{1}{\sqrt{a\pi\hbar}} \int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{1}{2} (e^{i\frac{\pi x}{a}} + e^{-i\frac{\pi x}{a}}) e^{-ipx/\hbar} dx \\
&= \frac{1}{2\sqrt{a\pi\hbar}} \left(\frac{1}{i(\frac{\pi}{a} - p/\hbar)} \right) e^{i\frac{\pi x}{a} - ipx/\hbar} - \frac{1}{2\sqrt{a\pi\hbar}} \left(\frac{1}{i(\frac{\pi}{a} + p/\hbar)} \right) e^{-i\frac{\pi x}{a} - ipx/\hbar} \\
&= \frac{1}{\sqrt{a\pi\hbar}} \left(\frac{1}{(\frac{\pi}{a} - p/\hbar)} \right) \sin\left(\frac{\pi}{a} - p/\hbar\right) \frac{a}{2} + \frac{1}{\sqrt{a\pi\hbar}} \left(\frac{1}{(\frac{\pi}{a} + p/\hbar)} \right) \sin\left(\frac{\pi}{a} + p/\hbar\right) \frac{a}{2}
\end{aligned}$$

2.4

$$\begin{aligned}
1 &= \int_0^a A^2 x^2 (x-a)^2 dx = \int_0^a A^2 (x^2 (x^2 - 2ax + a^2)) dx \\
1 &= \int_0^a A^2 (x^4 - 2ax^3 + a^2 x^2) dx = A^2 \left[\frac{1}{5} a^5 - \frac{1}{2} a^5 + \frac{1}{3} a^5 \right] = \frac{1}{30} a^5 \\
\Rightarrow A &= \sqrt{\frac{30}{a^5}} \\
C &= \int_0^a \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} A x (x-a) dx = A \sqrt{\frac{2}{a}} \int_0^a \sin \frac{n\pi x}{a} x (x-a) dx \\
&= A \sqrt{\frac{2}{a}} \int_0^a x^2 \sin \frac{n\pi x}{a} dx - A \sqrt{2a} \int_0^a \sin \frac{n\pi x}{a} x dx \\
(1) &= A \sqrt{\frac{2}{a}} \frac{a}{n\pi} a^2 (-1)^{n+1} + A \sqrt{\frac{2}{a}} \frac{a}{n\pi} \int_0^a 2x \cos \frac{n\pi x}{a} dx
\end{aligned}$$

$$\begin{aligned}
(1)(2) &= A\sqrt{\frac{2}{a}} \frac{2a}{n\pi} \int_0^a x \cos \frac{n\pi x}{a} dx = A\sqrt{\frac{2}{a}} \frac{2a}{n\pi} \frac{a}{n\pi} x \sin \frac{n\pi x}{a} - A\sqrt{\frac{2}{a}} \frac{2a}{n\pi} \frac{a}{n\pi} \int_0^a \sin \frac{n\pi x}{a} dx \\
&= A\sqrt{\frac{2}{a}} \frac{2a}{n\pi} \frac{a}{n\pi} \frac{a}{n\pi} [(-1)^n - 1] = A\sqrt{2a} \frac{2a^2}{n^3\pi^3} [(-1)^n - 1] \\
(2) &= -A\sqrt{2a} \int_0^a \sin \frac{n\pi x}{a} x dx = -A\sqrt{2a} \frac{a}{n\pi} (-\cos \frac{n\pi x}{a}) x + 0 \\
&= A\sqrt{2a} \frac{a^2}{n\pi} (-1)^n \\
(1) + (1)(2) + (2) &= A\sqrt{2a} \frac{2a^2}{n^3\pi^3} [(-1)^n - 1] \\
n = 1 &\Rightarrow C = A\sqrt{2a} \frac{-4a^2}{\pi^3} \\
P_1 &= \frac{30}{a^5} \frac{32a^5}{\pi^6} = \frac{960}{\pi^6}
\end{aligned}$$

2.6

$$\begin{aligned}
\psi &= \begin{cases} e^{ikx} + Re^{-ikx} & x < 0 \\ Te^{ik'x} & x > 0 \end{cases} \\
-\frac{\hbar^2}{2m} \nabla^2 \psi - (V + E)\psi &= 0 \Rightarrow k = \sqrt{\frac{2m(V + E)}{\hbar^2}} \\
k' &= \sqrt{\frac{2mE}{\hbar^2}} \\
1 + R = T; k - kR = k'T &\Rightarrow R = \frac{k - k'}{k + k'} \Rightarrow |R|^2 = \frac{(k - k')^2}{(k + k')^2} \\
Tran &= 1 - \frac{(k - k')^2}{(k + k')^2} = \frac{4kk'}{(k + k')^2}
\end{aligned}$$

2.7,2.8,2.9

$$x = \frac{1}{\sqrt{2}}(a + a^\dagger)$$

$$\frac{1}{\sqrt{2}}(a + a^\dagger)|n\rangle = \frac{1}{\sqrt{2}}(\sqrt{n}|n-1\rangle + \sqrt{n+1}|n+1\rangle)$$

$$x^2 = \frac{1}{2}(a^2 + (2N+1) + a^{\dagger 2})$$

$$x^2|n\rangle = \frac{1}{2}(\sqrt{n}\sqrt{n-1}|n-2\rangle + (2n+1)|n\rangle + \sqrt{n+1}\sqrt{n+2}|n+2\rangle)$$

$$p = \frac{1}{\sqrt{2}i}(a - a^\dagger)$$

$$p|n\rangle = \frac{1}{\sqrt{2}i}(\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle)$$

$$p^2|n\rangle = -\frac{1}{2}(a^2 - (2N+1) + a^{\dagger 2})|n\rangle$$

$$= -\frac{1}{2}(\sqrt{n}\sqrt{n-1}|n-2\rangle - (2n+1)|n\rangle + \sqrt{n+1}\sqrt{n+2}|n+2\rangle)$$

$$\bar{p} = \bar{x} = 0$$

$$\bar{x}^2 = (n + \frac{1}{2})\frac{\hbar}{\mu\omega}, \bar{p}^2 = (n + \frac{1}{2})\hbar\omega\mu$$

$$\Delta x = \sqrt{(n + \frac{1}{2})\frac{\hbar}{\mu\omega}}, \Delta p = \sqrt{(n + \frac{1}{2})\hbar\omega\mu}$$

$$\Delta x \Delta p = (n + \frac{1}{2})\hbar$$

13 Oct 20 HW

Oct 20 homework: Proov 3 equation

$$1.[p_x, \psi(x)] = -i\hbar \frac{\partial \psi}{\partial x}$$

$$2.[p_x^2, \psi(x)] = -2i\hbar \frac{\partial \psi}{\partial x} p_x - \hbar^2 \frac{\partial^2 \psi}{\partial x^2}$$

$$3.[x, y(p(x))] = i\hbar \frac{\partial y(p(x))}{\partial p(x)}$$

proov now:

$$1:[p_x, \psi]\psi = p_x[\psi^2] - \psi p_x[\psi] = -i\hbar 2\psi \partial_x \psi + i\hbar \psi \partial_x \psi = -i\hbar \partial_x \psi \psi$$

$$2:[p_x^2, \psi] = -\hbar^2 \partial_x \partial_x \psi^2 + \hbar^2 \psi \partial_x \partial_x \psi = -2\hbar^2 \partial_x \psi \times p_x \psi - \hbar^2 \psi \partial_x \partial_x \psi = (-2\hbar^2 \partial_x \psi p_x - \hbar^2 \partial_x \partial_x \psi) \psi$$

$$3:[x, y(p(x))] = \Sigma_n C_n p(x)^n \text{ so } \Sigma_n C_n [x, p_x^n] = [x, y(p(x))] \text{ and } [x, p_x^n] \psi = x \times p_x^n \psi - p_x^n (x\psi) = i\hbar n p_x^{n-1} \psi = \Sigma_n C_n [x, p_x^n] = [x, \Sigma_n C_n p_x^n] = i\hbar \Sigma_n C_n n p_x^{n-1} \psi = i\hbar \partial_{p_x} y(p_x) \psi$$

$$p_x^n (x\psi) = p_x^{n-1} (c\psi + x \partial_x \psi) = c p_x^{n-1} \psi + p_x^{n-2} (c \partial_x \psi + x p_x^2 \psi) \text{ so } = n c p_x^{n-1} \psi +$$

$$p_x^n \psi \text{ and } c = -i\hbar$$

$$p_x [x\psi] = -i\hbar \psi + x p_x \psi$$

D O N E .

14 Oct 23HW part2

PROOF: $e^{a \frac{d}{dt}} \psi = \psi(x + a)$

$$e^{a \frac{d}{dt}} \psi = \sum_n \frac{a^n}{n!} \frac{d^n}{dt^n} \psi; \psi(x + y) = \sum_n \frac{\frac{d^n \psi}{dt^n} |_{y=0}}{n!} y^n \text{ consider } y=a: \sum_n \frac{\frac{d^n \psi(x)}{dt^n}}{n!} a^n = \sum_n \frac{a^n}{n!} \frac{d^n}{dt^n} \psi = e^{a \frac{d}{dt}} \psi$$

DONE.

15 Oct 28HW

A:

let $l_{\pm} = l_x \pm il_y$

PROOF:

(a). $[l_z, l_{\pm}] = \pm \hbar l_{\pm}$

(b). $l_{\pm} l_{\mp} = l^2 - l_z^2 \pm \hbar l_z$

(c). $[l_+, l_-] = 2\hbar l_z$

proof a:

$$[l_z, l_{\pm}] = [l_z, l_x] \pm i[l_z, l_y] = i\hbar l_y \pm i(-i\hbar l_x) = i\hbar l_y \pm \hbar l_x = \pm \hbar l_{\pm}$$

proof b:

$$l_{\pm} l_{\mp} = (l_x \pm il_y)(l_x \mp il_y) = l_x^2 \mp il_x l_y \pm il_y l_x + l_y^2 = l_x^2 + l_y^2 \pm i(l_y l_x - l_x l_y) = l^2 - l_z^2 \pm i[l_y, l_x] = l^2 - l_z^2 \pm \hbar l_z$$

proof c:

$$l_+ l_- - l_- l_+ = (l_x + il_y)(l_x - il_y) - (l_x - il_y)(l_x + il_y) = -il_x l_y + il_y l_x - il_x l_y + il_y l_x = 2i[l_y, l_x] = 2\hbar l_z$$

B:

PROOF:

(a) $p \times l + l \times p = 2i\hbar p$

(b) $i\hbar(p \times l - l \times p) = [l^2, p]$

proof a:

$$(p_1 l_2 - p_2 l_1)k + (p_2 l_3 - p_3 l_2)i + (p_3 l_1 - p_1 l_3)j + (l_1 p_2 - l_2 p_1)k + (l_2 p_3 - l_3 p_2)i + (l_3 p_1 - l_1 p_3)j =$$

$$([p_1, l_2] + [l_1, p_2])k + ([p_2, l_3] + [l_2, p_3])i + ([p_3, l_1] + [l_3, p_1])j$$

$$(i2\hbar p_3)k + (i2\hbar p_1)i + (i2\hbar p_2)j = i2\hbar \mathbf{p}$$

proof b:

$$i\hbar(p \times l - l \times p) = (p_1 l_2 - p_2 l_1 - l_1 p_2 + l_2 p_1)k + (p_2 l_3 - p_3 l_2 - l_2 p_3 + l_3 p_2)i + (p_3 l_1 - p_1 l_3 - l_3 p_1 + l_1 p_3)j$$

$$[l^2, p] = [l^2, p] = [l_x^2 + l_y^2 + l_z^2, p]$$

$$[l_x^2, p] = l_x [l_x, p] + [l_x, p] l_x = l_x (i\hbar(p_z j - p_y k)) + (i\hbar(p_z j - p_y k)) l_x = i\hbar(l_x p_z + p_z l_x)j - i\hbar(l_x p_y + p_y l_x)k$$

$$[l_y^2, p] = l_y [l_y, p] + [l_y, p] l_y = l_y (i\hbar(p_x k - p_z i)) + (i\hbar(p_x k - p_z i)) l_y = i\hbar(l_y p_x + p_x l_y)k - i\hbar(l_y p_z + p_z l_y)i$$

$$\begin{aligned}
[l_z^2, p] &= l_z[l_z, p] + [l_z, p]l_z = l_z i\hbar(p_y i - p_x j) + i\hbar(p_y i - p_x j)l_z = i\hbar(l_z p_y + \\
& p_y l_z)i - i\hbar(l_z p_x + p_x l_z)j \\
[l^2, p] &= i\hbar(l_y p_x + p_x l_y - l_x p_y - p_y l_x)k + i\hbar(l_z p_y + p_y l_z - l_y p_z - p_z l_y)i + \\
& i\hbar(l_x p_z + p_z l_x - l_z p_x - p_x l_z)j \\
& \text{so we have } i\hbar(p \times l - l \times p) = [l^2, p]
\end{aligned}$$

16 Oct 30 HW

$$\begin{aligned}
\psi_n &= A_n e^{-\alpha^2 x^2/2} H_n(\alpha x) \\
2.7 \\
(1) \\
H_n(\alpha x) &= \frac{1}{2\alpha x} (H_{n+1}(\alpha x) + 2n H_{n-1}(\alpha x)) \\
x\psi_n &= \frac{1}{2\alpha} (H_{n+1}(\alpha x) + 2n H_{n-1}(\alpha x)) A_n e^{-\alpha^2 x^2/2} \\
A_n &= \frac{1}{\sqrt{2n}} A_{n-1}; A_n = \sqrt{2n+2} A_{n+1} \\
\text{so we get: } x\psi_n &= \frac{1}{2\alpha} (A_n H_{n+1}(\alpha x) + 2n A_n H_{n-1}(\alpha x)) e^{-\alpha^2 x^2/2} \\
\frac{1}{2\alpha} (\sqrt{2(n+1)} A_{n+1}(\alpha x) + 2n \frac{1}{\sqrt{2n}} A_{n-1} H_{n-1}(\alpha x)) e^{-\alpha^2 x^2/2} \\
&= \frac{1}{2\alpha} (\sqrt{2(n+1)} \psi_{n+1} + \sqrt{2n} \psi_{n-1}) \\
(2) \\
x^2 \psi_n &= \frac{1}{2\alpha} (\sqrt{2(n+1)} x \psi_{n+1} + \sqrt{2n} x \psi_{n-1}) \\
x^2 \psi_n &= \frac{1}{2\alpha} (\sqrt{2(n+1)} [\frac{1}{2\alpha} (\sqrt{2(n+2)} \psi_{n+2} + \sqrt{2(n+1)} \psi_n)] + \sqrt{2n} [\frac{1}{2\alpha} (\sqrt{2n} \psi_n + \\
\sqrt{2(n-1)} \psi_{n-2})]) &= \frac{1}{4\alpha^2} [\sqrt{4(n+1)(n+2)} \psi_{n+2} + 2(n+1) \psi_n + 2n \psi_n + \sqrt{4n(n-1)} \psi_{n-2}] \\
&= \frac{1}{2\alpha^2} [\sqrt{(n+1)(n+2)} \psi_{n+2} + (2n+1) \psi_n + \sqrt{n(n-1)} \psi_{n-2}] \\
(3) \\
\bar{x} &= \int \psi_n^* x \psi_n = 0 \\
V &= \frac{1}{2} K x^2; \bar{V} = \int \psi_n^* V \psi_n = \frac{1}{2} K \int \psi_n^* x^2 \psi_n = \frac{1}{2} K \frac{1}{2\alpha^2} (2n+1) = \frac{K(2n+1)}{4\alpha^2} = \\
\frac{\omega(2n+1)\hbar}{4} &= \frac{E_n}{2} \\
2.8 \\
H'_n(\alpha z) &= 2n\alpha H_{n-1}(\alpha z) \\
\psi'_n &= A_n e^{-\alpha^2 x^2/2} 2n\alpha H_{n-1}(\alpha x) - \alpha^2 x A_n e^{-\alpha^2 x^2/2} [\frac{1}{2\alpha x} (H_{n+1} + 2n H_{n-1})] \\
&= 2n\alpha A_n e^{-\alpha^2 x^2/2} H_{n-1} - \frac{\alpha}{2} A_n e^{-\alpha^2 x^2/2} H_{n+1} - \alpha n A_n e^{-\alpha^2 x^2/2} H_{n-1} \\
&= \alpha n A_n e^{-\alpha^2 x^2/2} H_{n-1} - \frac{\alpha}{2} A_n e^{-\alpha^2 x^2/2} H_{n+1} \\
A_n &= \frac{1}{\sqrt{2n}} A_{n-1}; A_n = \sqrt{2n+2} A_{n+1} \\
&= \alpha \sqrt{\frac{n}{2}} \psi_{n-1} - \alpha \sqrt{\frac{n+1}{2}} \psi_{n+1} \\
(2) \\
\frac{d}{dx} \psi'_n &= \alpha [\sqrt{\frac{n}{2}} \psi'_{n-1} - \sqrt{\frac{n+1}{2}} \psi'_{n+1}] \\
\frac{d}{dx} \psi'_n &= \alpha [\sqrt{\frac{n}{2}} [\alpha \sqrt{\frac{n-1}{2}} \psi_{n-2} - \alpha \sqrt{\frac{n}{2}} \psi_n] - \sqrt{\frac{n+1}{2}} [\alpha \sqrt{\frac{n+1}{2}} \psi_n] - \alpha \sqrt{\frac{n+2}{2}} \psi_{n+2}] \\
\frac{d}{dx} \psi'_n &= \alpha^2/2 [\sqrt{(n-1)n} \psi_{n-2} - n \psi_n - (n+1) \psi_n + \sqrt{(n+1)(n+2)} \psi_{n+2}] \\
\frac{d}{dx} \psi'_n &= \alpha^2/2 [\sqrt{(n-1)n} \psi_{n-2} - (2n+1) \psi_n + \sqrt{(n+1)(n+2)} \psi_{n+2}] \\
(3) \\
\bar{p} &= \int \psi_n^* p \psi_n = 0
\end{aligned}$$

$$\begin{aligned}
\bar{T} &= \int \psi_n^* T \psi_n = \int \psi_n^* (-\hbar^2/(2m)) \frac{d^2}{dx^2} \psi_n = \frac{-\hbar^2}{2m} \int \psi_n^* \frac{d^2}{dx^2} \psi_n = \frac{-\hbar^2}{2m} [-\frac{\alpha^2}{2}(2n+1)] \\
&= \frac{\hbar\alpha^2}{4m}(2n+1) = \frac{\hbar\omega}{4}(2n+1) = \frac{1}{2}E_n \\
2.9 \\
\Delta x \Delta p &= \sqrt{x^2 p^2} = \sqrt{E_n m E_n / K} = \frac{E_n}{\omega} = \frac{\hbar}{2}(2n+1) \\
3.14 \\
l_z \text{ Eigenstate: } \psi_m &= \sqrt{\frac{1}{2\pi}} e^{im\phi} \\
\int \phi^* l_z \phi &= \int (l_z \phi)^* \phi \\
\int \phi^* [l_y, l_z] \phi &= \int \phi^* l_y l_z \phi - \phi^* l_z l_y \phi = \int \phi^* l_y \phi \bar{h} m - (l_z \phi)^* l_y \phi = \bar{h} m \bar{l}_y - \bar{h} m \bar{l}_y = 0 \\
&= i\hbar l_x \\
\int \phi^* [l_z, l_x] \phi &= \int \phi^* l_z l_x \phi - \phi^* l_x l_z \phi = \int (l_z \phi)^* l_x \phi - (l_x \phi)^* l_z \phi = \int \phi^* \bar{h} m l_x \phi - \bar{h} m \phi^* l_x \phi = 0 = i\hbar l_y
\end{aligned}$$

17 Nov 4 HW

3.15

$$\begin{aligned}
\psi &= Y_{lm} \text{ so } \bar{l}_x^2 = \int \psi^* l_x^2 \psi d\Omega \text{ and } l_x^2 = \frac{1}{i\hbar} [l_y, l_z] l_x = \frac{1}{i\hbar} (l_y l_z - l_z l_y) l_x = \\
&\frac{1}{i\hbar} (l_y l_z l_x - l_z l_y l_x) \text{ and we know } l_z l_x = l_x l_z + i\hbar l_y \text{ so } : \frac{1}{i\hbar} (l_y l_x l_z + i\hbar l_y^2 - l_z l_y l_x) = \\
&\frac{1}{i\hbar} l_y l_x l_z + l_y^2 - \frac{1}{i\hbar} l_z l_y l_x = l_x^2 \\
&\text{so } \bar{l}_x^2 = \int \psi^* (\frac{1}{i\hbar} l_y l_x l_z + l_y^2 - \frac{1}{i\hbar} l_z l_y l_x) \psi d\Omega = \bar{l}_y^2 + \int \psi^* \frac{1}{i\hbar} l_t l_x \bar{h} m \psi - \frac{1}{i\hbar} \bar{h} m \psi^* l_y l_x \psi d\Omega = \bar{l}_y^2 \\
&\text{so we have } \bar{l}_x^2 = \bar{l}_y^2 \text{ now} \\
&\text{we know } l^2 = l_x^2 + l_y^2 + l_z^2, \text{ so } \bar{l}^2 = 2\bar{l}_x^2 + \bar{l}_z^2 \\
&\bar{l}_x^2 = \frac{1}{2} (\int \psi^* l^2 \psi - \psi^* l_z^2 \psi d\Omega) = \frac{1}{2} (l(l+1)\hbar^2 - m^2\hbar^2) \\
&\text{so } (\Delta \bar{l}_x)^2 \text{ we have } : \Delta l_x = \sqrt{-\bar{l}_x^2 + \bar{l}^2} = \sqrt{\bar{l}_z^2} \text{ and thus } (\Delta l_x)^2 = |\bar{l}_z|^2 = \\
&\frac{1}{2} (l(l+1)\hbar^2 - m^2\hbar^2) \text{ so we have} \\
&(\Delta \bar{l}_x)^2 = \frac{1}{2} (l(l+1)\hbar^2 - m^2\hbar^2) = (\Delta \bar{l}_y)^2
\end{aligned}$$

3.16

$$\begin{aligned}
&\text{set } \psi = c_1 Y_{11} + c_2 Y_{20} \\
&l_z \psi = c_1 \hbar Y_{11} + c_2 \times 0 = c_1 \hbar Y_{11} \text{ so the possibility of status } 0\hbar \text{ is } |c_2|^2, \text{ and } \bar{h} \\
&\text{is 1, so } l_z = \hbar \text{ and the average } = |c_1|^2 \hbar \\
&l^2 \psi = c_1 \times 1 \times 2Y_{11} \hbar^2 + c_2 \times 2 \times 3Y_{20} \hbar^2 \text{ so } l^2 \text{ can be } 2\hbar^2 \text{ possibility is } |c_1|^2, \text{ and } \\
&6\hbar^2 \text{ possibility is } |c_2|^2.
\end{aligned}$$

18 Nov 6 HW

*3.18 Use The Taylor Expend Can Proof It Very Easily

$$\begin{aligned}
e^{-aA} B e^{aA} &= \sum \frac{-a^n A^n}{n!} B \sum \frac{a^m A^m}{m!} = \sum \sum \frac{-a^n A^n}{n!} B \frac{a^m A^m}{m!} = \sum_{m,n} \frac{-1^n a^{m+n} A^n B A^m}{n! m!} \\
&= \sum_{m,n} \frac{-1^n a^{m+n} A^n B A^m}{n! m!} = B - aAB + aBA - a^2ABA + a^2BA^2/2 + a^2A^2B/2... \\
&= B - a[A, B] - \frac{1}{2}a^2(A[A, B])... \text{ for any } m+n=i \text{ we can have} \\
&a^i \sum_{m+n=i} \frac{-1^n A^n B A^m}{n! m!} \text{ for } i=3 \text{ we can have: } a^3 \sum_{m+n=3} \frac{-1^n A^n B A^m}{n! m!} \\
&= a^3 [[A, [A, B]], A]/6 = -\frac{a^3}{3!} [A, [A, [A, B]]]
\end{aligned}$$

Now set for $i = 0, 1, 2, 3$ we set $F_i = \frac{-a}{i+1} [A, F_{i-1}] = \Sigma_{m+n=i} \frac{a^i (-1)^n A^n B A^m}{m! n!}$
Now consider for ant k satisfied F_k satisfied : $F_k = \frac{a^k}{k!} (-1)^k [A, [A, [A, \dots [A, B]]]] =$
 $\Sigma_{m+n=k} \frac{a^k (-1)^n A^n B A^m}{m! n!}$ and for $k+1$ we have:
 $-a/(k+1) [A, F_k] = \frac{a^{k+1}}{(k+1)!} (-1)^{k+1} [A, [A, \dots [A, B]]] = \Sigma_{m+n=k} \frac{a^{k+1} (-1)^n [A, A^n B A^m]}{m! n! (k+1)}$
and for $[A, A^n B A^m] = [A, A^n] B A^m + A^n [A, B A^m] = [A, A^n] B A^m + A^n [A, B] A^m +$
 $A^n B [A, A^m] = A^n [A, B] A^m = A^{n+1} B A^m - A^n B A^{m+1} = A^{n+1} B A^m + (-1) A^n B A^{m+1}$ so
 $\Sigma_{m+n=k} \frac{a^{k+1} (-1)^n (A^{n+1} B A^m + (-1) A^n B A^{m+1})}{m! n! (k+1)}$
 $\Sigma_{m+n=k} \frac{a^{k+1} (-1)^{n+1} A^{n+1} B A^m + a^{k+1} (-1)^n A^n B A^{m+1}}{m! n! (k+1)} = \Sigma_{m+n'=k+1} \frac{a^{k+1} (-1)^{n'} A^{n'} B A^m}{m! (n'-1)! (k+1)} +$
 $\Sigma_{m'+n=k+1} \frac{a^{k+1} (-1)^n A^n B A^{m'}}{(m'-1)! n! (k+1)} - \frac{a^{k+1} A B A^k}{(k+1)!} - \frac{a^{k+1} (-1)^k A^k B A}{(k+1)!} = \Sigma_{m+n'=k+1} n' \frac{a^{k+1} (-1)^{n'} A^{n'} B A^m}{m! (n')! (k+1)} +$
 $\Sigma_{m'+n=k+1} m' \frac{a^{k+1} (-1)^n A^n B A^{m'}}{(m')! n! (k+1)} + \frac{a^{k+1} A B A^k}{(k+1)!} - \frac{a^{k+1} (-1)^k A^k B A}{(k+1)!}$ 1 and 3 combine can
have a $m \neq 1$ term and 2 and 4 combine can have a $n \neq 1$ term : $\Sigma_{m+n'=k+1; m \neq 1} n' \frac{a^{k+1} (-1)^{n'} A^{n'} B A^m}{m! (n')! (k+1)} +$
 $\Sigma_{m'+n=k+1; n \neq 1} m' \frac{a^{k+1} (-1)^n A^n B A^{m'}}{(m')! n! (k+1)}$ so we get satisfied for full $k+1$ terms : $m=0, 1, 2, \dots, k$
total $k+1$, and we set $m+n=k+1$, we get : $\Sigma \frac{a^{k+1} (-1)^n A^n B A^m}{m! n!}$
PROOF DONE.

19 Nov 13

4.4: proof: $-\hbar^2 \frac{d^2}{dt^2} \bar{A} = \overline{[A, H], H}$

PROOF: $\frac{d}{dt} \bar{C} = \frac{1}{i\hbar} [\bar{C}, H]$ and let $\bar{C} = \frac{d}{dt} \bar{A} = \frac{1}{i\hbar} \overline{[A, H]}$ so

$$\frac{d}{dt} \frac{d}{dt} \bar{A} = -\frac{1}{\hbar^2} \overline{[A, H], H}$$

PROVED

4.5: proof: $\frac{d}{dt} \bar{A} = 0$ $\bar{A}$ is under a stationary state.

PROVE: $\frac{d}{dt} \bar{A} = \frac{1}{i\hbar} \overline{[A, H]} = \frac{1}{i\hbar} \int \psi^* A H \psi - \psi^* H A \psi = \frac{1}{i\hbar} (E \int \psi^* A \psi - E \int \psi^* A \psi) =$

0

DONE

20 Nov 18

homework:

4.2:

(a)

$$H = h(q_1) + h(q_2)$$

$$q_1, q_2 = \phi_1, \phi_1; \phi_1, \phi_2; \phi_1, \phi_3; \phi_2, \phi_2; \phi_2, \phi_3; \phi_3, \phi_3.$$

6 kinds

(b)

$$H = h(q_1) + h(q_2)$$

$$q_1, q_2 = \phi_1, \phi_2, \phi_2, \phi_1; \phi_1, \phi_3; \phi_3, \phi_1; \phi_2, \phi_3; \phi_3, \phi_2.$$

3 kinds

(c)

$$3 \times 3 = 9$$

9 kinds

4.3:

(a)

$$3^3 = 27$$

27 kinds

(b)

$$\phi_{q_1}, \phi_{q_2}, \phi_{q_3} :$$

$$1, 2, 3$$

1 kind

(c)

$$\phi_{q_1}, \phi_{q_2}, \phi_{q_3} :$$

$$1, 1, 1; 1, 1, 2; 1, 1, 3; 1, 2, 2; 1, 2, 3;$$

$$1, 3, 3; 2, 2, 2; 2, 2, 3; 2, 3, 3; 3, 3, 3;$$

10 kinds

4.8:

proof:

$$\left(\frac{dF}{dt}\right)_{nn'} = i\omega_{nn'} F_{nn'}, \omega_{nn'} = (E_n - E_{n'})/\hbar \quad (1)$$

prove:

$$\begin{aligned}
\frac{dF}{dt} &= \frac{1}{i\hbar} [F(t), H] \\
\left(\frac{dF}{dt}\right)_{nn'} &= \frac{1}{i\hbar} \int \phi_n F H \phi_{n'} - \phi_n H F \phi_{n'} \\
&= \frac{1}{i\hbar} (E_{n'} \int \phi_n F \phi_{n'} - E_n \int \phi_n F \phi_{n'}) \\
&= \frac{1}{i\hbar} (E_{n'} - E_n) \int \phi_n F \phi_{n'} \\
&= -\frac{1}{i\hbar} \hbar \omega_{nn'} F_{nn'} \\
&= i\omega_{nn'} F_{nn'}
\end{aligned}$$

21 Nov 20

4.7: proof:

$$\frac{\partial E_n}{\partial \lambda} = \langle \psi_n | \frac{\partial H}{\partial \lambda} | \psi_n \rangle \quad (2)$$

PROVE:

$$\begin{aligned} E_n \psi_n &= \hat{H} \psi_n \\ \psi_{na}^* E_n \psi_n^a &= E_n = \psi_{na}^* \hat{H} \psi_n^a \\ \frac{\partial E_n}{\partial \lambda} &= \frac{\partial}{\partial \lambda} (\psi_{na}^* \hat{H} \psi_n^a) \\ &= \psi_{na}^* \frac{\partial \hat{H}}{\partial \lambda} \psi_n^a = \langle \psi_n | \frac{\partial \hat{H}}{\partial \lambda} | \psi_n \rangle \end{aligned}$$

PROVED

22 Nov 25 NOT THIS !!!!!HW is untill page 10

$\psi = c_1 Y_{11} + c_2 Y_{20}$ try to solve all result and probablility of $\hat{l}_x \psi$
SOLVE:

23 Nov 27 HW

homework

7.1:

IF: $A^2 = B^2 = C^2 = 1; BC - CB = iA$.

PROVE: (a) $AB + BA = AC + CA = 0$; (b) B, C in image of A.

PROVE:

(a):

$$iAB + iBA = BCB - CBB + BBC - BCB = BBC - CBB = 0$$

$$iAC + iCA = BCC - CBC + CBC - CCB = 0$$

(b)/home/yuri/books/output/1/1758960262251.jpg

$$AB + BA = 0; \phi_i^* AB \phi_i + \phi^* iBA \phi_j = 0$$

$$\lambda_I \phi_i^* B \phi_j + \phi^{*i} B \lambda_J \phi_j = \lambda_I B_j^i + \lambda_J B_j^i = (\lambda_I + \lambda_J) B_j^i = 0;$$

$$B^2 = 1 \neq 0, B_i^i = 0, B_j^j \neq 0; \lambda_I + \lambda_J = 0; \lambda_1 = a; \lambda_2 = -a$$

$$A = \begin{bmatrix} a & 0 \\ 0 & -a \end{bmatrix}; B = \begin{bmatrix} 0 & b_1 \\ b_2 & 0 \end{bmatrix}; C = \begin{bmatrix} 0 & c_1 \\ c_2 & 0 \end{bmatrix}$$

$$BB = \begin{bmatrix} b_1 b_2 & 0 \\ 0 & b_1 b_2 \end{bmatrix}; b_1 = \frac{1}{b_2}; B = \begin{bmatrix} 0 & b \\ \frac{1}{b} & 0 \end{bmatrix}; C = \begin{bmatrix} 0 & c \\ \frac{1}{c} & 0 \end{bmatrix}$$

$$BC = \begin{bmatrix} \frac{b}{c} & 0 \\ 0 & \frac{c}{b} \end{bmatrix}; CB = \begin{bmatrix} \frac{c}{b} & 0 \\ 0 & \frac{b}{c} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}; B = \begin{bmatrix} 0 & b \\ \frac{1}{b} & 0 \end{bmatrix}; C = \begin{bmatrix} 0 & c \\ \frac{1}{c} & 0 \end{bmatrix}$$

$$BC - CB = iA; ia = \frac{b}{c} - \frac{c}{b};$$

7.2:

$$A^2 = 0, AA^\dagger + A^\dagger A = 1, B = A^\dagger A.$$

PROVE: (a) $B^2 = B$; (b) in image of B to solve A.

(a)

$$B^2 = A^\dagger AA^\dagger A = A^\dagger A(1 - AA^\dagger) = A^\dagger A = B$$

PROVED

(b)

$$B_j^i = \text{diag}(\lambda_1, \lambda_2)$$

$$B^2 = B; B = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}; \lambda_1^2 = \lambda_1, \lambda_2^2 = \lambda_2$$

$$\text{WLOG set } B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; A^\dagger A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, AA^\dagger = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}; A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}; A^2 = \begin{bmatrix} a^2 + bc & ab + bd \\ ca + bd & cb + d^2 \end{bmatrix} = 0; a = d = 0$$

$$bb^* = 0, cc^* = 1; b = 0, c = \sqrt{-k^2 + 1} + ki$$

24 Dec 3 HW

I:8.1:

(a):

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0; \lambda = \pm 1; \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \times \vec{v} = 0$$

$$\vec{v} = \begin{bmatrix} k \\ k \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\lambda = -1; \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \times \vec{v}_1 = 0; \vec{v}_1 = \begin{bmatrix} k \\ -k \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{bmatrix}$$

(b):

$$\phi_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}; \phi_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{bmatrix}$$

$$\psi_i^j = \psi_i^a \phi_a^{*j}; \psi_1^1 = \frac{1}{\sqrt{2}}; \psi_1^2 = \frac{1}{\sqrt{2}}; \psi_2^1 = \frac{1}{\sqrt{2}}; \psi_2^2 = \frac{-1}{\sqrt{2}}$$

$$\phi^a = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \times \psi^a$$

(c):

$$S^{-1} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$S\sigma_x S^{-1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

II:8.2:

$$\begin{aligned}
\sigma \cdot n &= \sigma_x n_x + \sigma_y n_y + \sigma_z n_z = \sigma_x \sin\theta \cos\phi + \sigma_y \sin\theta \sin\phi + \sigma_z \cos\theta \\
\phi^{*i} \sigma \cdot n \phi_j &= \sigma_x^i \sin\theta \cos\phi + \sigma_y^i \sin\theta \sin\phi + \sigma_z^i \cos\theta \\
&= \begin{bmatrix} \cos\theta & \sin\theta \cos\phi - i \sin\theta \sin\phi \\ \sin\theta \cos\phi + i \sin\theta \sin\phi & -\cos\theta \end{bmatrix} \\
&= \begin{bmatrix} \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta \end{bmatrix}
\end{aligned}$$

III:CALCULATE: $[\sigma_x, \sigma_z^n]; [\sigma_x, e^{\sigma_z}]$

$$\begin{aligned}
\sigma_z^n &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}^n = \begin{bmatrix} 1 & 0 \\ 0 & -1^n \end{bmatrix}; \sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & -1^n \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & -1^n \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 & -1^n \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -1^n & 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 & -1^n - 1 \\ 1 + -1^{n+1} & 0 \end{bmatrix}
\end{aligned}$$

25 Dec 4

8.2:

$$\begin{aligned}
\sigma \cdot n &= \sigma_x n_x + \sigma_y n_y + \sigma_z n_z \\
&= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \sin\theta \cos\phi + \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \times \sin\theta \sin\phi + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \cos\theta \\
&= \begin{bmatrix} \cos\theta & \sin\theta \cos\phi - i \sin\theta \sin\phi \\ \sin\theta \cos\phi + i \sin\theta \sin\phi & -\cos\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta \end{bmatrix}
\end{aligned}$$

8.3:

$$\begin{aligned}
\overline{S_x^2} &= \phi^* S_x^2 \phi = \phi^* \frac{1}{4} (S_+ + S_-)(S_+ + S_-) \phi \\
&= [\phi^* (S_+^2 + S_+ S_- + S_- S_+ + S_-^2) \phi] = [\phi^* (S_+ S_-) \phi] = [\phi^* \frac{\hbar^2}{4} \phi] = \frac{\hbar^2}{4} \\
\overline{S_x} &= \phi^* \frac{1}{2} (S_+ + S_-) \phi = 0; \sqrt{\overline{S_x^2} - 0} = \frac{\hbar}{2} \\
(\overline{\Delta S_x})^2 &= \frac{\hbar^2}{4} \\
\overline{S_y^2} &= \phi^* - 1 \times \frac{1}{4} (S_+ - S_-)(S_+ - S_-) \phi = -\frac{1}{4} \phi^* (S_+^2 - S_+ S_- - S_- S_+ + S_-^2) \phi \\
&= -\frac{1}{4} \phi^* (-\hbar^2 \phi) = \frac{\hbar^2}{4} \\
\overline{S_y} &= \phi^* S_y \phi = \frac{1}{2i} \phi^* (S_+ - S_-) \phi = 0 \\
\overline{\Delta S_y^2} &= (\sqrt{\overline{S_y^2} - \overline{S_y}^2})^2 = \frac{\hbar^2}{4}
\end{aligned}$$

26 Dec 10

I:Calc:

$$\hat{H} = g\sigma_{x1}\sigma_{x2} \quad ; \quad \psi_0^{ab} = \frac{1}{\sqrt{2}}\chi_{\frac{1}{2}}^a\chi_{\frac{1}{2}}^b + \frac{1}{\sqrt{2}}\chi_{\frac{1}{2}}^a\chi_{-\frac{1}{2}}^b$$

Solve:

use base $\{\chi_{00}, \chi_{10}, \chi_{11}, \chi_{1-1}\} = \{\phi_0, \phi_1, \phi_2, \phi_3\}$

$$= \frac{1}{\sqrt{2}}(\chi_1\chi_0 - \chi_0\chi_1); \frac{1}{\sqrt{2}}(\chi_1\chi_0 + \chi_0\chi_1); \chi_1\chi_1; \chi_0\chi_0$$

$$\hat{H}_j^i = \phi_b^{*i} \hat{H}_a^b \phi_j^a$$

H_j^i	ϕ_0	ϕ_1	ϕ_2	ϕ_3
ϕ_0^*	$-g$	0	0	0
ϕ_2^*	0	g	0	0
ϕ_3^*	0	0	0	g
ϕ_4^*	0	0	g	0

$$\hat{H}_0^0 = \frac{1}{2}\phi_0^*g(\chi_0\chi_1 - \chi_1\chi_0) = \frac{1}{2}g(\chi_1^*\chi_0^* - \chi_0^*\chi_1^*)(\chi_0\chi_1 - \chi_1\chi_0) = -g$$

$$\hat{H}_1^0 = \frac{1}{2}g\phi_0^*(\chi_0\chi_1 + \chi_1\chi_0) = \frac{1}{2}g(\chi_1^*\chi_0^* - \chi_0^*\chi_1^*)(\chi_0\chi_1 + \chi_1\chi_0) = 0$$

$$\hat{H}_2^0 = \hat{H}_3^0 = 0$$

$$\hat{H}_1^1 = \frac{1}{2}g\phi_1^*(\chi_0\chi_1 + \chi_1\chi_0) = \frac{1}{2}g(\chi_1\chi_0 + \chi_0\chi_1)(\chi_0\chi_1 + \chi_1\chi_0) = g$$

$$\hat{H}_2^1 = \hat{H}_3^1 = 0$$

$$\hat{H}_2^2 = g\phi_2^*\hat{H}\phi_2 = 0$$

$$\hat{H}_3^2 = g\phi_2^*\hat{H}\phi_3 = g$$

$$\hat{H}_3^3 = 0$$

$$\begin{vmatrix} -g-\lambda & 0 & 0 & 0 \\ 0 & g-\lambda & 0 & 0 \\ 0 & 0 & -\lambda & g \\ 0 & 0 & g & -\lambda \end{vmatrix} = 0; \quad \text{so } \lambda = (\lambda^2 - g^2)(\lambda^2 - g^2) = 0$$

$$v_0^a = \phi_0; v_1^a = \phi_1; v_2^a = \frac{1}{\sqrt{2}}(\phi_2 + \phi_3); v_3^a = \frac{1}{\sqrt{2}}(\phi_2 - \phi_3); \lambda_0 = -g; \lambda_1 = g; \lambda_2 = g$$

$$; \lambda_3 = -g$$

$$\psi_0^{ab} = \frac{1}{\sqrt{2}}\phi_2^a + \frac{1}{2}(\phi_0 + \phi_1) = \frac{1}{2}v_0 + \frac{1}{2}v_1 + \frac{1}{2}(v_3 + v_2)$$

II:8.8:

$$\psi_0 = \chi_1 \chi_0 \text{ and the Hamiltonian : } \hat{H} = A s_1 \cdot s_2.$$

Solve:

$$\hat{H} = \frac{A}{4} \hbar^2 \sigma_1 \sigma_2 = \frac{A}{4} \hbar^2 (\sigma_{1x} \sigma_{2x} + \sigma_{1y} \sigma_{2y} + \sigma_{1z} \sigma_{2z})$$

$$\text{bases are : } \chi_{00} = \frac{1}{\sqrt{2}} (\chi_1 \chi_0 - \chi_0 \chi_1); \chi_{10} = \frac{1}{\sqrt{2}} (\chi_1 \chi_0 + \chi_0 \chi_1); \chi_{11} = \chi_1 \chi_1; \chi_{1-1} = \chi_0 \chi_0$$

$$\hat{H}_j^i = \phi_a^{*i} \hat{H}_b^a \phi_j^b$$

$\hat{H}_j^i$	ϕ_0	ϕ_1	ϕ_2	ϕ_3
ϕ_0^*	$-\frac{3}{4} A \hbar^2$	0	0	0
ϕ_1^*	0	$\frac{1}{4} A \hbar^2$	0	0
ϕ_2^*	0	0	$\frac{1}{4} A \hbar^2$	0
ϕ_3^*	0	0	0	$\frac{1}{4} A \hbar^2$

$$\hat{H}_0^a = \frac{A}{4} \hbar^2 \frac{3}{\sqrt{2}} (\chi_0 \chi_1 - \chi_1 \chi_0) = -\frac{3}{4} A \hbar^2 \chi_{00}$$

$$\hat{H}_0^0 = -\frac{3}{4} A \hbar^2$$

$$\hat{H}_1^a = \frac{1}{4} A \hbar^2 \frac{1}{\sqrt{2}} (\chi_0 \chi_1 + \chi_1 \chi_0) = \frac{1}{4} A \hbar^2 \chi_{10}$$

$$\hat{H}_1^1 = \frac{1}{4} A \hbar^2$$

$$\hat{H}_2^a = \frac{1}{4} A \hbar^2 (\chi_0 \chi_0 - \chi_0 \chi_0 + \chi_1 \chi_1) = \frac{1}{4} A \hbar^2 \chi_{11}$$

$$\hat{H}_2^2 = \frac{1}{4} A \hbar^2$$

$$\hat{H}_3^a = \frac{1}{4} A \hbar^2 (\chi_1 \chi_1 - \chi_1 \chi_1 + \chi_0 \chi_0) = \frac{1}{4} A \hbar^2 \chi_{1-1}$$

$$\hat{H}_3^3 = \frac{1}{4} A \hbar^2$$

$$\psi_0 = \frac{1}{\sqrt{2}} (\chi_{00} + \chi_{10})$$

$$\begin{aligned} \psi_0(t) &= \frac{1}{\sqrt{2}} \chi_{00} e^{i \frac{3}{4} A \hbar t} + \frac{1}{\sqrt{2}} \chi_{10} e^{-i \frac{1}{4} A \hbar t} = \frac{1}{2} e^{i \frac{3}{4} \hbar A t} (\chi_1 \chi_0 - \chi_0 \chi_1) + \frac{1}{2} e^{-i \frac{1}{4} \hbar A t} (\chi_1 \chi_0 + \chi_0 \chi_1) \\ &= e^{i \frac{1}{4} \hbar A t} \left[\frac{1}{2} e^{i \frac{1}{2} \hbar A t} (\chi_1 \chi_0 - \chi_0 \chi_1) + \frac{1}{2} e^{-\frac{1}{2} \hbar A t} (\chi_1 \chi_0 + \chi_0 \chi_1) \right] \end{aligned}$$

$$= e^{i\frac{1}{4}A\hbar t} [\cos(\frac{1}{2}\hbar At)\chi_1\chi_0 + \frac{1}{i}\sin(\frac{1}{2}\hbar At)\chi_0\chi_1]$$

(a) :

$$= \cos^2(\frac{1}{2}\hbar At)$$

(b) :

$$= 0$$

(c) :

$$= \frac{1}{2}; = \frac{1}{2}$$

(d) :

$$< s_{1x} > = (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t}) s_{1x} (\frac{1}{\sqrt{2}}\chi_{00} e^{i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10} e^{-i\frac{1}{4}A\hbar t}) = 0$$

$$< s_{1y} > = (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t}) s_{1y} (\frac{1}{\sqrt{2}}\chi_{00} e^{i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10} e^{-i\frac{1}{4}A\hbar t}) = 0$$

$$\begin{aligned} < s_{1z} > &= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t}) s_{1z} (\frac{1}{\sqrt{2}}\chi_{00} e^{i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10} e^{-i\frac{1}{4}A\hbar t}) \\ &= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t}) \frac{1}{2}\hbar (\frac{1}{2}e^{i\frac{3}{4}\hbar At}(\chi_1\chi_0 + \chi_0\chi_1) + \frac{1}{2}e^{-i\frac{1}{4}\hbar At}(\chi_1\chi_0 - \chi_0\chi_1)) \\ &= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t}) \frac{1}{2}\hbar (\frac{1}{\sqrt{2}}e^{i\frac{3}{4}\hbar At}\chi_{10} + \frac{1}{\sqrt{2}}e^{-i\frac{1}{4}\hbar At}\chi_{00}) \\ &= \frac{1}{2}e^{i\hbar At} + \frac{1}{2}e^{-i\hbar At} = \frac{1}{2}\hbar \cos(\hbar At) \\ < s_{2x} > &= < s_{2y} > = 0 \\ < s_{2z} > &= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t}) s_{2z} (\frac{1}{\sqrt{2}}\chi_{00} e^{i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10} e^{-i\frac{1}{4}A\hbar t}) \\ &= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t}) s_{2z} (\frac{1}{2}e^{i\frac{3}{4}A\hbar t}(\chi_1\chi_0 - \chi_0\chi_1) + \frac{1}{2}e^{-i\frac{1}{4}A\hbar t}(\chi_1\chi_0 + \chi_0\chi_1)) \\ &= (\frac{1}{\sqrt{2}}\chi_{00}^* e^{-i\frac{3}{4}A\hbar t} + \frac{1}{\sqrt{2}}\chi_{10}^* e^{i\frac{1}{4}A\hbar t}) s_{2z} (\frac{1}{\sqrt{2}}e^{i\frac{3}{4}A\hbar t}(-\chi_{10}) + \frac{1}{\sqrt{2}}e^{-i\frac{1}{4}A\hbar t}(-\chi_{00})) \\ &= -\frac{1}{2}\hbar \cos(A\hbar t) \end{aligned}$$

27 Dec 11 HW

I:8.5:

(a):Prove

$$e^{i\lambda\sigma_z} = \cos\lambda + i\sigma_z \sin\lambda$$

Proof.

$$\begin{aligned} e^{i\lambda\sigma_z} &= \sum \frac{(i\lambda)^n}{n!} \sigma_z^n; \cos\lambda = \sum \frac{(-1)^n}{2n!} \lambda^{2n} \\ \sin\lambda &= \sum \frac{(-1)^n}{2n+1!} \lambda^{2n+1} \\ \sigma_z &= \begin{bmatrix} 1 & 0 \\ 0 & (-1)^n \end{bmatrix}; e^{i\lambda\sigma_z} = \sum \begin{bmatrix} \frac{(i\lambda)^n}{n!} & 0 \\ 0 & \frac{(-i\lambda)^n}{n!} \end{bmatrix} \\ \cos\lambda + i\sigma_z \sin\lambda &= \begin{bmatrix} \cos\lambda + i\sin\lambda & 0 \\ 0 & \cos\lambda - i\sin\lambda \end{bmatrix} \\ &= \begin{bmatrix} e^{i\lambda} & 0 \\ 0 & e^{-i\lambda} \end{bmatrix} = e^{i\lambda\sigma_z} \end{aligned}$$

□

(b):Prove

$$e^{i\sigma \cdot A} = \cos A + i\sigma \cdot n \sin A, A = An, A = |A|$$

Proof.

$$\begin{aligned} e^{i\sigma \cdot A} &= e^{i\sigma \cdot nA} = \sum \frac{(iA)^n}{n!} (\sigma \cdot n)^n \\ \cos A + i\sigma \cdot n \sin A \\ \sigma \cdot n &= n_x \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + n_y \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} + n_z \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} \\ - &> \begin{bmatrix} \cos A + i\sin A n_z & i\sin A (n_x - in_y) \\ i\sin A (n_x + in_y) & \cos A - i\sin A n_z \end{bmatrix} \\ e^{i\sigma \cdot nA} &= \sum \frac{(iA)^n}{n!} \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix}^n \\ &= \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} \times \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \sum \frac{-1^n A^{2n}}{2n!} I + \sum \frac{-1^n \times iA^{2n+1}}{2n+1} \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} = \cos AI + i\sin A \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} \\ &= \cos A + i\sigma \cdot n \sin A \end{aligned}$$

□

(c): Prove

$$Tre^{i\sigma \cdot A} = 2\cos A$$

Proof.

$$\begin{aligned} e^{i\sigma \cdot A} &= \cos A I + i \sin A \begin{bmatrix} n_z & n_x - i n_y \\ n_x + i n_y & -n_z \end{bmatrix} \text{ the trace is :} \\ &= 2\cos A + 0 = 2\cos A \end{aligned}$$

□

II:8.6:
Prove:

$$e^{i\lambda\sigma_z}\sigma_x e^{-i\lambda\sigma_z} = \sigma_x \cos 2\lambda - \sigma_y \sin 2\lambda$$

Proof.

$$\begin{aligned} & (\cos\lambda + i\sigma_z \sin\lambda)\sigma_x(\cos\lambda - i\sigma_z \sin\lambda) \\ &= \begin{bmatrix} \cos\lambda + i\sin\lambda & 0 \\ 0 & \cos\lambda - i\sin\lambda \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \cos\lambda - i\sin\lambda & 0 \\ 0 & \cos\lambda + i\sin\lambda \end{bmatrix} \\ &= \begin{bmatrix} \cos\lambda + i\sin\lambda & 0 \\ 0 & \cos\lambda - i\sin\lambda \end{bmatrix} \begin{bmatrix} 0 & \cos\lambda + i\sin\lambda \\ \cos\lambda - i\sin\lambda & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & \cos^2\lambda - \sin^2\lambda + 2i\cos\lambda\sin\lambda \\ \cos^2\lambda - \sin^2\lambda - 2i\cos\lambda\sin\lambda & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & e^{i2\lambda} \\ e^{-i2\lambda} & 0 \end{bmatrix} \\ \sigma_x \cos 2\lambda - \sigma_y \sin 2\lambda &= \begin{bmatrix} 0 & \cos 2\lambda + i\sin 2\lambda \\ \cos 2\lambda - i\sin 2\lambda & 0 \end{bmatrix} = \begin{bmatrix} 0 & e^{i2\lambda} \\ e^{-i2\lambda} & 0 \end{bmatrix} \end{aligned}$$

□

III:Backboard Question:

$$H = \frac{\hbar^2}{4} g \sigma_{1x} \sigma_{2x}; \psi = \frac{1}{\sqrt{2}} [\chi_1 \chi_0 + \chi_1 \chi_1]$$

Solve.

$$\text{the eigenstate: } \phi_1 = \frac{1}{\sqrt{2}} [\chi_1 \chi_0 + \chi_0 \chi_1]; \phi_2 = \frac{1}{\sqrt{2}} [\chi_1 \chi_0 - \chi_0 \chi_1];$$

$$\phi_3 = \frac{1}{\sqrt{2}} [\chi_1 \chi_1 + \chi_0 \chi_0]; \phi_4 = \frac{1}{\sqrt{2}} [\chi_1 \chi_1 - \chi_0 \chi_0]$$

$$\lambda_1 = 1; \lambda_2 = -1; \lambda_3 = 1; \lambda_4 = -1$$

$$\psi_0 = \frac{1}{2} [\phi_1 + \phi_2] + \frac{1}{2} [\phi_3 + \phi_4]$$

$$\phi(t) = \frac{1}{2} \phi_1 e^{-it/\hbar} + \frac{1}{2} \phi_2 e^{it/\hbar} + \frac{1}{2} \phi_3 e^{-it/\hbar} + \frac{1}{2} \phi_4 e^{it/\hbar}$$

$$\phi^*(t) = \frac{1}{2} \phi_1^* e^{it/\hbar} + \frac{1}{2} \phi_2^* e^{-it/\hbar} + \frac{1}{2} \phi_3^* e^{it/\hbar} + \frac{1}{2} \phi_4^* e^{-it/\hbar}$$

$$\langle \sigma_{x1} \rangle = \phi^*(t) \sigma_z \frac{1}{2} [e^{-it/\hbar} \frac{1}{\sqrt{2}} (\chi_1 \chi_0 + \chi_0 \chi_1 + \chi_1 \chi_1 + \chi_0 \chi_0) + e^{it/\hbar} \frac{1}{\sqrt{2}} (\chi_1 \chi_0 - \chi_0 \chi_1 + \chi_1 \chi_1 - \chi_0 \chi_0)]$$

$$= \phi^*(t) \frac{1}{2\sqrt{2}} [e^{-it/\hbar} (\chi_0 \chi_0 + \chi_1 \chi_1 + \chi_0 \chi_1 + \chi_1 \chi_0) + e^{it/\hbar} (\chi_0 \chi_0 - \chi_1 \chi_1 + \chi_0 \chi_1 - \chi_1 \chi_0)]$$

$$= \phi^*(t) [e^{-it/\hbar} \frac{1}{2} (\phi_1 + \phi_3) - \frac{1}{2} e^{it/\hbar} (\phi_2 + \phi_4)] = 0$$

Solved.

28 Dec 16 HW

I:8.9:

Solve:

$$\psi = \chi_1$$

$$H = \frac{eB\hbar}{2mc}\sigma_x, \text{ eigenstate is } : \phi_1 = \frac{1}{\sqrt{2}}[\chi_1 + \chi_0]; \phi_2 = \frac{1}{\sqrt{2}}[\chi_1 - \chi_0]$$

$$\lambda_1 = \frac{eB\hbar}{2mc}; \lambda_2 = -\frac{eB\hbar}{2mc}$$

$$\psi = \frac{1}{\sqrt{2}}(\phi_1 + \phi_2)$$

$$\psi(t) = \frac{1}{\sqrt{2}}\phi_1 e^{-i\frac{eB}{2mc}t} + \frac{1}{\sqrt{2}}\phi_2 e^{i\frac{eB}{2mc}t}$$

$$\langle s_x \rangle = \frac{1}{\sqrt{2}}[\phi_1^* e^{iKt} + \phi_2^* e^{-iKt}] \frac{1}{\sqrt{2}} \frac{\hbar}{2} [\phi_1 e^{-iKt} - \phi_2 e^{-iKt}] = \frac{\hbar}{4}(0) = 0$$

$$\langle s_y \rangle = \frac{1}{\sqrt{2}}[\phi_1^* e^{iKt} + \phi_2^* e^{-iKt}] \frac{\hbar}{2} \frac{1}{\sqrt{2}} [-e^{-iKt} i \phi_2 + e^{iKt} i \phi_1] = \frac{\hbar}{4} i (e^{2iKt} - e^{-2iKt}) = \frac{-\hbar}{2} \sin(2Kt)$$

$$\langle s_z \rangle = \frac{1}{\sqrt{2}}[\phi_1^* e^{iKt} + \phi_2^* e^{-iKt}] \frac{\hbar}{2} \frac{1}{\sqrt{2}} [e^{-iKt} \phi_2 + e^{iKt} \phi_1] = \frac{\hbar}{4} (e^{i2Kt} + e^{-i2Kt}) = \frac{\hbar}{2} \cos(2Kt)$$

II:8.10

Solve

$$\phi_0 = \frac{\sqrt{\alpha}}{\pi^{\frac{1}{4}}} e^{-\alpha^2 x^2 / 2}$$

$$\phi_1 = \frac{\sqrt{2\alpha}}{\pi^{\frac{1}{4}}} \alpha x e^{-\alpha^2 x^2 / 2}$$

$$\phi_2 = \frac{1}{\pi^{\frac{1}{4}}} \sqrt{\frac{\alpha}{2}} (2\alpha^2 x^2 - 1) e^{-\alpha^2 x^2 / 2}$$

two particles without spin can be (wave function must be symmetry):

$$\phi_{02}\phi_{01} : \psi = \frac{\alpha}{\sqrt{\pi}} e^{-\frac{\alpha^2}{2}(x_1^2 + x_2^2)} E = \hbar\omega$$

$$\phi_{11}\phi_{02}, \phi_{01}\phi_{12} : \psi = \frac{\alpha^2}{\sqrt{\pi}} [(x_1 + x_2) e^{-\frac{\alpha^2}{2}(x_1^2 + x_2^2)}] E = 2\hbar\omega$$

$$\phi_{11}\phi_{12}, \phi_{01}\phi_{22}, \phi_{21}\phi_{02} : \psi = \frac{1}{\sqrt{2}} \frac{\alpha}{\sqrt{2\pi}} [2\alpha^2(x_1^2 + x_2^2 - 2) e^{-\frac{\alpha^2}{2}(x_1^2 + x_2^2)}];$$

$$\phi = \frac{2\alpha^3}{\sqrt{\pi}} x_1 x_2 e^{-\frac{\alpha^2}{2}(x_1^2 + x_2^2)} E = 3\hbar\omega$$

two particle with spin can be (wavefunction must be anti-symmetry):

eigenstate of Hamiltonian:

$$\phi_0 \chi_{sm_s}$$

$$\phi_{01}\phi_{02}\chi_{00} : \psi = \frac{\alpha}{\sqrt{\pi}} e^{-\frac{\alpha^2}{2}(x_1^2+x_2^2)} \chi_{00}$$

the second class

$$\psi = \frac{1}{\sqrt{2}}[\phi_{01}\phi_{12} + \phi_{11}\phi_{02}]\chi_{00}; \psi = \frac{1}{\sqrt{2}}[\phi_{01}\phi_{12} - \phi_{11}\phi_{02}]\chi_{1M}$$

the third class

$$\psi = [\phi_{11}\phi_{12}]\chi_{00}; \psi = \frac{1}{\sqrt{2}}[\phi_{01}\phi_{22} + \phi_{21}\phi_{02}]\chi_{00}; \psi = \frac{1}{\sqrt{2}}[\phi_{01}\phi_{22} - \phi_{21}\phi_{02}]\chi_{1M}$$

III:5.6

Calculate

(a):

$$\begin{aligned} Pdr|_r &= \int |\psi_{rlm}|^2 r^2 \sin\theta dr d\theta d\phi = \int R_r^2 Y_{lm}^2 r^2 \sin\theta dr d\theta d\phi = R^2 r^2 dr \int |Y_{lm}|^2 \sin\theta d\theta d\phi \\ &= R^2 r^2 dr = \chi^2 dr \end{aligned}$$

$$\text{so when } P \text{ is the biggest then : } \frac{dP}{dr} = 2\chi \frac{d\chi}{dr} = 0$$

$$\Rightarrow \frac{d(r^{l+1}e^{-\beta r})}{dr} = (l+1)r^l e^{-\beta r} - \beta r^{l+1} e^{-\beta r} = 0$$

$$\Rightarrow r = \frac{l+1}{\beta} = \frac{n}{\beta} = n^2$$

consider the dimension of length: $L = \frac{\hbar^2}{\mu e^2} = \alpha$ but the atom have Z electronics so:

$$\Rightarrow r = n^2 \frac{\alpha}{Z}$$

(b):

$$\langle r \rangle = \int \psi_{rlm}^* r \psi_{rlm} d\Omega$$

$$\text{use the virial theorem } \frac{\overline{p^2}}{2m} = \frac{1}{2} \overline{r \cdot \nabla V}$$

$$E = T + V = \frac{p^2}{2m} + \frac{1}{r} = \frac{\mu Z^2 e^4}{2\hbar^2 n^2} = \frac{Z^2 e^2}{2\alpha n^2}$$

$$\overline{T} = -\frac{1}{2}\overline{V}; \overline{E} = \frac{1}{2}\overline{V} \Rightarrow \frac{\overline{1}}{r} = \frac{Z}{\alpha n^2}$$

From 5.9 we know an equation:

$$\frac{\lambda+1}{n^2} \langle r^\lambda \rangle - (2\lambda+1) \frac{a_0}{Z} \langle r^{\lambda-1} \rangle + \frac{\lambda}{4} [(2l+1)^2 - \lambda^2] \frac{a_0^2}{Z^2} \langle r^{\lambda-2} \rangle = 0$$

so we get :

$$\begin{aligned} \frac{2}{n^2} \langle r \rangle - 3 \frac{a_0}{Z} + \frac{1}{4} [(2l+1)^2 - 1] \frac{a_0^2}{Z^2} \langle \frac{1}{r} \rangle &= 0 \\ \Rightarrow \langle r \rangle &= (n^2 + \frac{n}{2}) \frac{\alpha}{Z} \end{aligned}$$

(c):

As the same put $\langle r \rangle$ into eq.1 can get

$$\langle r^2 \rangle = n^2(n+1)(n+1) \left(\frac{\alpha}{2}\right)^2$$

$$\text{so delta equal to : } \left(\frac{n^3}{2} + \frac{n^2}{4}\right)^{\frac{1}{2}} \frac{\alpha}{Z}$$

29 HW Dec 19

Prove and Calculate:

From

$$[a, a^\dagger] = 1$$

Calculate:

$$(1) : [a, a^\dagger a]$$

$$(2) : [a, (a^\dagger a)^n]$$

Calculation:

$$(1) : [a, a^\dagger a] = a^\dagger [a, a] + [a, a^\dagger] a = a$$

$$(2) : [a, (a^\dagger a)^2] = (a^\dagger a) a + a (a^\dagger a)$$

$$[a, (a^\dagger a)^3] = (a^\dagger a)^2 a + a^\dagger a a a^\dagger a + a (a^\dagger a)^2$$

Now set the n-1 satisfied:

$$[a, (a^\dagger a)^{n-1}] = (a^\dagger a)^{n-2} a + (a^\dagger a)^{n-3} a a^\dagger a + \dots + a^\dagger a a (a^\dagger a)^{n-3} + a (a^\dagger a)^{n-2}$$

For n we have:

$$\begin{aligned} [a, (a^\dagger a)^n] &= (a^\dagger a)^{n-1} a + [a, (a^\dagger a)^{n-1}] a^\dagger a \\ &= (a^\dagger a)^{n-1} a + (a^\dagger a)^{n-2} a a^\dagger a + (a^\dagger a)^{n-3} a (a^\dagger a)^2 + \dots + a (a^\dagger a)^{n-1} \end{aligned}$$

DONE.

30 HW Dec 23

I:(1) Calculate

$$\psi_0 = |n\rangle \quad \text{calculate : } \langle \hat{x} \rangle, \langle \hat{p} \rangle, \langle \hat{x}^2 \rangle, \langle \hat{p}^2 \rangle, \Delta x, \Delta p \quad (3)$$

Solve:

$$(a) : \psi^* x \psi$$

$$\begin{aligned} x_b^a &= \frac{1}{\sqrt{2}} \sqrt{\frac{\hbar}{\omega\mu}} (a + a^\dagger) \\ \Rightarrow \psi_0^* \frac{1}{\sqrt{2}} \sqrt{\frac{\hbar}{\omega\mu}} (\sqrt{n}|n-1\rangle + \sqrt{n+1}|n+1\rangle) &= 0 \end{aligned}$$

$$(b) : \psi_0^* p \psi_0$$

$$\begin{aligned} \hat{p} &= \frac{1}{i\sqrt{2}} (a - a^\dagger) \sqrt{\hbar\omega\mu} \\ \Rightarrow \psi_0^* \frac{1}{i\sqrt{2}} \sqrt{\hbar\omega\mu} (\sqrt{n}|n-1\rangle + \sqrt{n+1}|n+1\rangle) &= 0 \end{aligned}$$

$$(c) : x^2 = \frac{1}{2} \frac{\hbar}{\omega\mu} (a^2 + aa^\dagger + a^\dagger a + a^{\dagger 2})$$

$$\begin{aligned} \psi_0^* \frac{1}{2} \frac{\hbar}{\omega\mu} (\sqrt{n(n-1)}|n-2\rangle + (2n+1)|n\rangle + \sqrt{(n+1)(n+1)}|n+2\rangle) \\ = \frac{1}{2} \frac{\hbar}{\omega\mu} (2n+1) \end{aligned}$$

$$(d) : p^2 = -\frac{1}{2} (\hbar\omega\mu) (a^2 - aa^\dagger - a^\dagger a + a^{\dagger 2})$$

$$\begin{aligned} \Rightarrow \psi^* \left(-\frac{1}{2} \hbar\omega\mu \right) (\sqrt{n(n-1)}|n-2\rangle - (2n+1)|n\rangle + \sqrt{(n+1)(n+1)}|n+2\rangle) \\ = \frac{1}{2} \hbar\omega\mu (2n+1) \end{aligned}$$

$$(e) : \Delta x = \sqrt{\langle x^2 \rangle - \bar{x}^2} = \sqrt{\frac{1}{2} \frac{\hbar}{\omega\mu} (2n+1)}$$

$$\Delta p = \sqrt{\langle p^2 \rangle - \bar{p}^2} = \sqrt{\frac{1}{2} \hbar\omega\mu (2n+1)}$$

(2) Calculate:

$$\psi_0 = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \Delta x \text{ and } \Delta p$$

Solve:

$$\begin{aligned} \bar{x} &= \psi^* \frac{1}{\sqrt{2}} \sqrt{\frac{\hbar}{\omega\mu}} (a + a^\dagger) \psi \\ &\Rightarrow \psi^* \frac{1}{\sqrt{2}} \sqrt{\frac{\hbar}{\omega\mu}} \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle + \sqrt{2}|2\rangle) = \frac{1}{\sqrt{2}} \sqrt{\frac{\hbar}{\omega\mu}} \end{aligned}$$

$$\begin{aligned} \bar{p} &= \psi^* \frac{1}{i\sqrt{2}} \sqrt{\hbar\omega\mu} (a - a^\dagger) \psi \\ &\Rightarrow \psi^* \frac{1}{i\sqrt{2}} \sqrt{\hbar\omega\mu} \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle - \sqrt{2}|2\rangle) = 0 \end{aligned}$$

$$\begin{aligned} p^2 &= -\frac{1}{2}(\hbar\omega\mu)(a^2 - aa^\dagger - a^\dagger a + a^{\dagger 2}) \\ &\Rightarrow \psi^* \left(-\frac{1}{2}(\hbar\omega\mu)\right)(a^2 - 2\hat{N} - 1 + a^{\dagger 2})\psi \\ &\Rightarrow \psi^* \left(-\frac{1}{2}\hbar\omega\mu\right) \frac{1}{\sqrt{2}} (-|0\rangle + \sqrt{2}|2\rangle - 3|1\rangle + \sqrt{6}|3\rangle) \\ &\Rightarrow \frac{1}{2} \left(-\frac{1}{2}\hbar\omega\mu\right)(-4) = \hbar\omega\mu \end{aligned}$$

$$\begin{aligned} x^2 &= \frac{1}{2} \frac{\hbar}{\omega\mu} (a^2 + aa^\dagger + a^\dagger a + a^{\dagger 2}) \\ &\Rightarrow \psi^* \left(\frac{1}{2} \frac{\hbar}{\omega\mu}\right) (a^2 + 2\hat{N} + 1 + a^{\dagger 2})\psi \\ &\Rightarrow \psi^* \left(\frac{\hbar}{2\omega\mu}\right) \frac{1}{\sqrt{2}} (|0\rangle + \sqrt{2}|2\rangle + 2|1\rangle + \sqrt{6}|3\rangle + |1\rangle) \\ &\Rightarrow \frac{1}{2} \frac{\hbar}{2\omega\mu} (4) = \frac{\hbar}{\omega\mu} \end{aligned}$$

$$\Delta p \Delta x = \sqrt{(\sqrt{\langle x^2 \rangle} - \bar{x})^2 (\sqrt{\langle p^2 \rangle} - \bar{p})^2} = \sqrt{(\sqrt{\frac{\hbar}{\omega \mu} - \frac{1}{2} \frac{\hbar}{\omega \mu}})^2 \sqrt{(\hbar \omega \mu - 0)^2}}$$

$$\Rightarrow \sqrt{\frac{\hbar}{2 \omega \mu}} \hbar \omega \mu = \frac{1}{\sqrt{2}} \hbar$$

31 Dec 29 HW

9.4:

The bases are $|1-1\rangle; |10\rangle; |11\rangle$

Set as ψ_1, ψ_2, ψ_3

$$l_x^i = \langle \psi^{*i} | l_x | \psi_j \rangle$$

$$l_x = \frac{1}{2}(l_+ + l_-)$$

$$l_1^1 = \psi^{*1} l_x \psi_1$$

$$\Rightarrow \psi^{*1} \frac{1}{2}(\sqrt{2}\psi_2) = 0$$

$$l_2^1 = \psi^{*1} l_x \psi_2 \Rightarrow \psi^{*1} \frac{1}{2}(\sqrt{2}\psi_1 + \sqrt{2}\psi_3) = \frac{\sqrt{2}}{2}$$

$$l_3^1 = \psi^{*1} l_x \psi_3 = \psi^{*1} \frac{1}{2}\sqrt{2} = 0$$

$$l_2^2 = \psi^{*2} l_x \psi_2 = \psi^{*2} \frac{1}{2}\sqrt{2}\psi_1 + \sqrt{2}\psi_3 = 0$$

$$l_3^2 = \psi^{*2} l_x \psi_3 = \psi^{*2} \frac{1}{2}\sqrt{2} = \frac{\sqrt{2}}{2}$$

$$l_3^3 = 0$$

$$l = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$$0 = \begin{vmatrix} -\lambda & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & -\lambda & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & -\lambda \end{vmatrix}$$

$$\lambda_1 = -1, \lambda_2 = 0, \lambda_3 = 1$$

$$v_1 = \frac{1}{2}\psi_1 - \frac{\sqrt{2}}{2}\psi_2 + \frac{1}{2}\psi_3; v_2 = \frac{\sqrt{2}}{2}\psi_1 - \frac{\sqrt{2}}{2}\psi_3; v_3 = \frac{1}{2}\psi_1 + \frac{\sqrt{2}}{2}\psi_2 + \frac{1}{2}\psi_3$$

32 Jan 6 HW

I:

$$V(x) = \begin{cases} A \cos \frac{\pi x}{a} & 0 \leq x \leq a \\ \infty & x < 0, x > a \end{cases}$$

Solve:

(1): $A \cos \frac{\pi x}{a}$ is .

(2):

$$V(x) = \begin{cases} 0 & 0 \leq x \leq a \\ \infty & x < 0, x > a \end{cases}$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi + V\psi = E\psi$$

$$\begin{cases} -\frac{\hbar^2}{2\mu} \nabla^2 \psi = E\psi & 0 \leq x \leq a \\ \psi = 0 & x < 0, x > a \end{cases}$$

$$\frac{d^2\psi}{dx^2} - \frac{2\mu E}{\hbar^2} \psi = 0$$

$$\psi = A \cos\left(\sqrt{\frac{2\mu E}{\hbar^2}} x + \phi\right) \quad 0 \leq x \leq a$$

$$\psi(0) = \psi(a) = 0 \Rightarrow \phi = \frac{\pi}{2}$$

$$\psi = A \sin\left(\sqrt{\frac{2\mu E}{\hbar^2}} x\right) \Rightarrow n\pi = \sqrt{\frac{2\mu E}{\hbar^2}} a$$

$$\psi = A \sin \frac{n\pi}{a} x$$

$$1 = A^2 \int_0^a \sin^2 \frac{n\pi}{a} x dx = A^2 \int_0^a \frac{1}{2} - \frac{1}{2} \cos \frac{2n\pi}{a} x dx$$

$$1 = A^2 \frac{1}{2} a \Rightarrow A = \sqrt{\frac{2}{a}}$$

$$E = \frac{n^2 \pi^2 \hbar^2}{2\mu a^2}$$

(3):

$$\begin{aligned}
\hat{H}^{(0)} &= -\frac{\hbar^2}{2\mu}\nabla^2; \hat{H}' = A\cos\frac{\pi x}{a} \quad 0 \leq x \leq a \\
\psi_n &= \sqrt{\frac{2}{a}}\sin\frac{n\pi}{a}x; E'_n = \int_0^a \psi_n^* A\cos\frac{\pi x}{a} \psi_n dx \\
&= \frac{A}{a} \int_0^a \cos\frac{\pi x}{a} - \cos\frac{2n\pi x}{a} \cos\frac{\pi x}{a} dx \\
&= \frac{A}{a} \left[\sin\frac{\pi a}{a} - 0 \right] - \frac{A}{a} \int_0^a \cos\frac{2n\pi x}{a} \cos\frac{\pi x}{a} dx \\
&= 0 \\
E_n^{(2)} &= \Sigma_m \frac{|H'_{mn}|^2}{E_n^{(0)} - E_m^{(0)}} \quad ; H'_{mn} = \frac{2}{a} \int_0^a \sin\frac{n\pi x}{a} H' \sin\frac{m\pi x}{a} dx \\
&\Rightarrow A \frac{2}{a} \int_0^a \sin\frac{n\pi x}{a} \cos\frac{\pi x}{a} \sin\frac{m\pi x}{a} dx \\
&= -\frac{A}{a} \int_0^a \cos\frac{\pi x}{a} \left[\cos\frac{\pi x(n+m)}{a} - \cos\frac{\pi x(n-m)}{a} \right] dx \\
&= -\frac{A}{a} \int_0^a \cos\frac{\pi x}{a} \cos\frac{\pi x(n+m)}{a} - \cos\frac{\pi x}{a} \cos\frac{\pi x(n-m)}{a} dx \\
&= -\frac{A}{a} \int_0^a \frac{1}{2} \left[\cos\frac{\pi x(m+n+1)}{a} + \cos\frac{\pi x(1-n-m)}{a} \right] - \\
&\quad \frac{1}{2} \left[\cos\frac{\pi x(1+n-m)}{a} + \cos\frac{\pi x(1-n+m)}{a} \right] dx \\
&\int_0^a \cos\frac{\pi x k}{a} dx = \frac{a}{k\pi} [\sin(\pi k) - 0] = 0 \quad k \neq 0 \\
&\int_0^a \cos\frac{\pi x k}{a} dx = a \quad k = 0 \\
&\int_0^a \cos\frac{\pi x k}{a} dx = a\delta(k) \\
&\Rightarrow -\frac{A}{a} \left\{ \frac{1}{2} [a\delta(m+n+1) + a\delta(1-n-m)] - \frac{1}{2} [a\delta(1+n-m) + a\delta(1-n+m)] \right\} \\
&= H'_{mn} \\
&= -\frac{A}{2} [\delta(m+n+1) + \delta(1-n-m) - \delta(1+n-m) - \delta(1-n+m)] \\
|H'_{mn}|^2 &= \frac{A^2}{4} [\delta(m+n+1) - \delta(n+1)\delta(m) - \delta(m+1)\delta(n) + \delta(1-n-m) \\
&\quad - \delta(m-1)\delta(n) - \delta(n-1)\delta(m) + \delta(1+n-m) \\
&\quad - \delta(n+1)\delta(m) - \delta(m-1)\delta(n) + \delta(1-n+m) - \delta(m+1)\delta(n) - \delta(n-1)\delta(m)]
\end{aligned}$$

$$\begin{aligned}
|H'_{mn}|^2 &= \frac{A^2}{4} [\delta(m+n+1) + \delta(1-n-m) + \delta(1-n+m) + \delta(1+n-m) \\
&\quad - 2\delta(n+1)\delta(m) - 2\delta(m-1)\delta(n) - 2\delta(m+1)\delta(n) - 2\delta(n-1)\delta(m)] \\
E_n^{(2)} &= \Sigma_m \frac{|H'_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}; E_n^{(0)} = \frac{n^2 \pi^2 \hbar^2}{2\mu a^2} \\
E_n^{(2)} &= \Sigma_m \frac{2\mu a^2}{\pi^2 \hbar^2} \frac{|H'_{mn}|^2}{n^2 - m^2} \\
|H'_{m1}|^2 &= \frac{A^2}{4} [\delta(m+2) + \delta(2-m)] \quad m > 0 \\
E_1^{(2)} &= \frac{2\mu a^2}{\pi^2 \hbar^2} \frac{A^2}{4} \left[\frac{1}{1-4} \right] = -\frac{1}{6} \frac{\mu a^2 A^2}{\pi^2 \hbar^2} \\
E_1 &= \frac{\pi^2 \hbar^2}{2\mu a^2} - \frac{1}{6} \frac{\mu a^2 A^2}{\pi^2 \hbar^2}
\end{aligned}$$

II:

$$H = \begin{bmatrix} 1 & c & 0 \\ c & 3 & 0 \\ 0 & 0 & c-2 \end{bmatrix}$$

(1):

$$\begin{vmatrix} 1-\lambda & c & 0 \\ c & 3-\lambda & 0 \\ 0 & 0 & c-2-\lambda \end{vmatrix} = 0 \Rightarrow (c-2-\lambda)((1-\lambda)(3-\lambda) - c^2) = 0$$

$$\lambda^2 - 4\lambda - c^2 + 3 = 0 \Rightarrow \lambda = \frac{4 \pm \sqrt{16 - 4(3 - c^2)}}{2}$$

$$\lambda_1 = c-2; \lambda_2 = 2 + \sqrt{1+c^2}; \lambda_3 = 2 - \sqrt{1+c^2}$$

(2):

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{bmatrix} + \begin{bmatrix} 0 & c & 0 \\ c & 0 & 0 \\ 0 & 0 & c \end{bmatrix} = \hat{H}^{(0)} + \hat{H}'$$

$$E_n^{(1)} = \hat{H}'_{nn} \Rightarrow E_1^{(1)} = 0, E_2^{(1)} = 0, E_3^{(1)} = c$$

$$E_n^{(2)} = \Sigma_m \frac{|H'_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}$$

$$E_1^{(2)} = \frac{c^2}{-2} + 0, E_2^{(2)} = \frac{c^2}{2} + 0, E_3^{(2)} = 0$$

$$E_1 = 1 + 0 - \frac{c^2}{2}, E_2 = 3 + \frac{c^2}{2}, E_3 = -2 + c$$

(3):

$$c = 0$$